Analysis of Factors Associated with Poor Response to Nutritional Treatment in Severely Malnourished Children Aged 6–59 Months Without Edema and With Complications Treated (Phase 1).

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1. Methodology

Descriptive analyses were conducted to characterize the cohort and the non-response to the nutritional treatment. Continuous variables were summarized using medians and interquartile ranges (25th - 75th percentile) due to their non-normal distribution, whereas categorical variables were summarized as counts and percentages (%).

Differences between groups (success and failure) were assessed using the Wilcoxon rank-sum test or Kruskal–Wallis test for continuous variables and Pearson's chi-square or Fisher's exact test for categorical variables, as appropriate.

Univariable logistic regression analyses were performed to estimate crude associations between independent variables and the dependent variable. Variables considered clinically relevant or statistically associated with the non-response to the nutritional treatment at $\underline{p} < 0.20$ were included in a multivariable logistic regression.

Prior to model building, pairwise correlations among independent variables were examined to identify highly correlated predictors and assess potential multicollinearity. Pearson correlation coefficients were computed for continuous variables (or Spearman's rank correlations for non-normally distributed variables). Variables exhibiting strong correlations ($|r| \geq 0.70$) were carefully evaluated to avoid redundancy. Variance Inflation Factors (VIFs) were subsequently calculated in the multivariable model, with VIF values > 5 indicating potential multicolliniarity.

Model selection was guided by likelihood ratio tests and the principle of parsimony. A stepwise selection procedure was used to identify the most parsimonious model (based on the Akaike Information Criterion (AIC)) by retaining variables that improved model fit while excluding predictors that contributed little additional information.

Adjusted odds ratios (aORs) with 95% confidence intervals (CIs) were reported. Statistical significance in the final model was defined as a two-sided $\underline{p} < 0.05$. A variable was significantly associated with the risk of nutritional treatment failure if: i) the odds ratio (OR) was > 1 with a lower limit of the confidence interval (CI) > 1; or ii) OR<1 with upper CI limit <1; and p-value was <0.05.

All analyses were conducted using R (version 4.3.0).

2. Study population

The analytic sample comprised 1,342 children aged 6-59 months treated for severe acute malnutrition during Phase 1, of whom 666 (49.6%) met the definition of non-response to treatment and 676 (50.4%) were classified as responders. The near-even split between groups provided balanced statistical power for bivariate and multivariable comparisons. Missingness was non-trivial and variable across covariates, ranging from negligible (12 missing for locality) to substantial (933 missing for breastfeeding status, 877 for food intolerance, 825 for death-in-phase, 770 for vaccination status), reflecting differential data capture across admission types rather than a uniform pattern of loss.

3. Descriptive Characteristics

The cohort was predominantly young, with a median age of 15 months (Q1-Q3: 10-22) and the large majority (81.6%) falling in the 6-24 month band rather than the 25-59 month band (18.4%). Sex distribution was balanced (55.2% male, 44.8% female). Most children resided in rural areas (75.6% vs. 24.4% urban). Vaccination coverage among those with data was relatively high (76.6% up to date). New admissions accounted for the majority of cases (86.2%), with readmissions comprising 13.8%. Comorbid infections were frequent: gastroenteritis (70.5%), pneumonia (59.3%), severe malaria (40.0%), and other/severe anemia (27.6% each) were the most prevalent, while tuberculosis (0.6%) and HIV (1.4%) were rare. Overall mortality during Phase 1 (among children with recorded outcome) was 15.9%, and 2.5% of children died on day 1. Type of discharge among those with data showed 66.3% recovered/transferred, 17.8% absconded, and 15.9% died.

4. Bivariate Analysis

Table 1: Association between explanatory variables and outcome at phase 1.

	Non-response to treatment in Phase				
Characteristic	**FALSE** N = 676[1]	**TRUE** N = 666[1]	**Overall** N = 1,342[1]	**p-value**[23]	**q-value**[4]
Age (Month)				0.051	0.071
Median (Q1 ; Q3)	15 (11.0 ; 23)	15 (10.0 ; 21)	15 (10.0 ; 22)		
Min ; Max	6 ; 59.0	6 ; 59.0	6 ; 59.0		
Age group				0.354	0.413
[25;59]	131 (19.4%)	116 (17.4%)	247 (18.4%)		
[6;25[	545 (80.6%)	550 (82.6%)	1,095 (81.6%)		
Gender				0.935	0.991
Male	374 (55.3%)	367 (55.1%)	741 (55.2%)		
Female	302 (44.7%)	299 (44.9%)	601 (44.8%)		
Locality				**0.004****	0.007
Urban	141 (21.0%)	184 (27.9%)	325 (24.4%)		
Rural	529 (79.0%)	476 (72.1%)	1,005 (75.6%)		
Missing values	6	6	12		
Vaccination status				0.495	0.541

Characteristic	Non-response to treatment in Phase				
	FALSE N = 676[1]	**TRUE** N = 666[1]	**Overall** N = 1,342[1]	**p-value**[2,3]	**q-value**[4]
Up to date	237 (77.7%)	201 (75.3%)	438 (76.6%)		
Out of date	68 (22.3%)	66 (24.7%)	134 (23.4%)		
Missing values	371	399	770		
Breastfeeding status	554 (94.7%)	562 (95.7%)	1,116 (95.2%)	0.404	0.456
Missing values	91	79	170		
Type admission				**0.022***	0.037
New admission	572 (88.4%)	547 (84.0%)	1,119 (86.2%)		
Readmission	75 (11.6%)	104 (16.0%)	179 (13.8%)		
Missing values	29	15	44		
Incident edema (P1)	0 (0.0%)	21 (3.2%)	21 (1.6%)	**<0.001*****	<0.001
WHZ error proportion (P1)				**<0.001*****	<0.001
Median (Q1 ; Q3)	1 (0.0 ; 1)	1 (0.0 ; 1)	1 (0.0 ; 1)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
Missing values	12	19	31		
Tube feeding duration (P1)				**<0.001*****	<0.001
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 1)	0 (0.0 ; 0)		
Min ; Max	0 ; 4.0	0 ; 19.0	0 ; 19.0		
Phase length (P1)				**<0.001*****	<0.001
Median (Q1 ; Q3)	3 (2.0 ; 3)	6 (5.0 ; 8)	4 (2.0 ; 6)		
Min ; Max	1 ; 4.0	1 ; 33.0	1 ; 33.0		
Nutritional treatment (P1)				**<0.001*****	<0.001
F75	356 (52.7%)	184 (27.6%)	540 (40.2%)		
F75F	245 (36.2%)	172 (25.8%)	417 (31.1%)		
F75-F75F/F75F-F75	48 (7.1%)	209 (31.4%)	257 (19.2%)		
Others	27 (4.0%)	101 (15.2%)	128 (9.5%)		
F75 to F75F transition (P1)	0 (0.0%)	240 (36.0%)	240 (17.9%)	**<0.001*****	<0.001
Family meals proportion (P1)				**0.029***	0.043
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
Food intolerance proportion (P1)				**<0.001*****	<0.001
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 1)	0 (0.0 ; 1)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 frequency below target (P1)				**0.002****	0.004

Characteristic	Non-response to treatment in Phase				
	FALSE N = 676[1]	**TRUE** N = 666[1]	**Overall** N = 1,342[1]	**p-value**[2,3]	**q-value**[4]
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 frequency above target (P1)				**0.024***	0.038
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 1.0	0 ; 1.0		
F75 quantity below target (P1)				**<0.001*****	<0.001
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 quantity above target (P1)				0.977	0.991
Median (Q1 ; Q3)	0 (0.0 ; 1)	0 (0.0 ; 1)	0 (0.0 ; 1)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 adherence below target (P1)				**<0.001*****	<0.001
Median (Q1 ; Q3)	0 (0.0 ; 1)	0 (0.0 ; 1)	0 (0.0 ; 1)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 adherence above target (P1)				**<0.001*****	<0.001
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
Morbidity burden				**<0.001*****	0.001
Median (Q1 ; Q3)	2 (1.0 ; 3)	2 (2.0 ; 3)	2 (1.0 ; 3)		
Min ; Max	0 ; 5.0	0 ; 6.0	0 ; 6.0		
Shock	8 (1.3%)	26 (4.0%)	34 (2.7%)	**0.003****	0.005
Missing values	53	18	71		
Gastroenteritis	385 (61.8%)	511 (78.9%)	896 (70.5%)	**<0.001*****	<0.001
Missing values	53	18	71		
Pneumonia	338 (54.3%)	416 (64.2%)	754 (59.3%)	**<0.001*****	<0.001
Missing values	53	18	71		
Tuberculosis	2 (0.3%)	6 (0.9%)	8 (0.6%)	0.288	0.348
Missing values	53	18	71		
Digestive candidiasis	54 (8.7%)	115 (17.7%)	169 (13.3%)	**<0.001*****	<0.001
Missing values	53	18	71		
HIV	6 (1.0%)	12 (1.9%)	18 (1.4%)	0.180	0.225
Missing values	53	18	71		
Severe malaria	290 (42.9%)	247 (37.1%)	537 (40.0%)	**0.030***	0.043

Characteristic	Non-response to treatment in Phase				
	FALSE N = 676[1]	**TRUE** N = 666[1]	**Overall** N = 1,342[1]	**p-value**[23]	**q-value**[4]
Neurological malaria	23 (3.7%)	24 (3.7%)	47 (3.7%)	0.991	0.991
Missing values	53	18	71		
Severe anemia	158 (25.4%)	193 (29.8%)	351 (27.6%)	0.078	0.101
Missing values	53	18	71		
Other anemia	158 (25.4%)	193 (29.8%)	351 (27.6%)	0.078	0.101
Missing values	53	18	71		
Death (P1)	0 (0.0%)	82 (30.5%)	82 (15.9%)	**<0.001*** **	<0.001
Missing values	428	397	825		
Death on day 1 (P1)	0 (0.0%)	33 (5.0%)	33 (2.5%)	**<0.001*** **	<0.001
Missing values	12	0	12		
Type of discharge (P1)				**<0.001*** **	<0.001
Recovered/Transferred	186 (75.0%)	157 (58.4%)	343 (66.3%)		
Absconded	62 (25.0%)	30 (11.2%)	92 (17.8%)		
Death	0 (0.0%)	82 (30.5%)	82 (15.9%)		
Missing values	428	397	825		

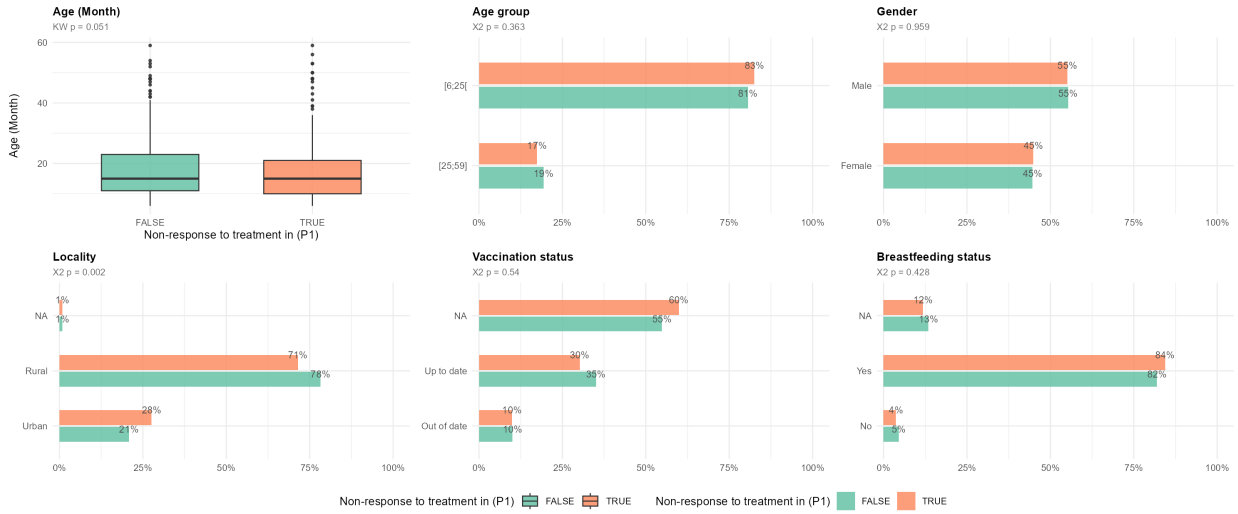
[1] n (%)

[2] Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test; Fisher's Exact Test for Count Data with simulated p-value
(based on 2000 replicates)

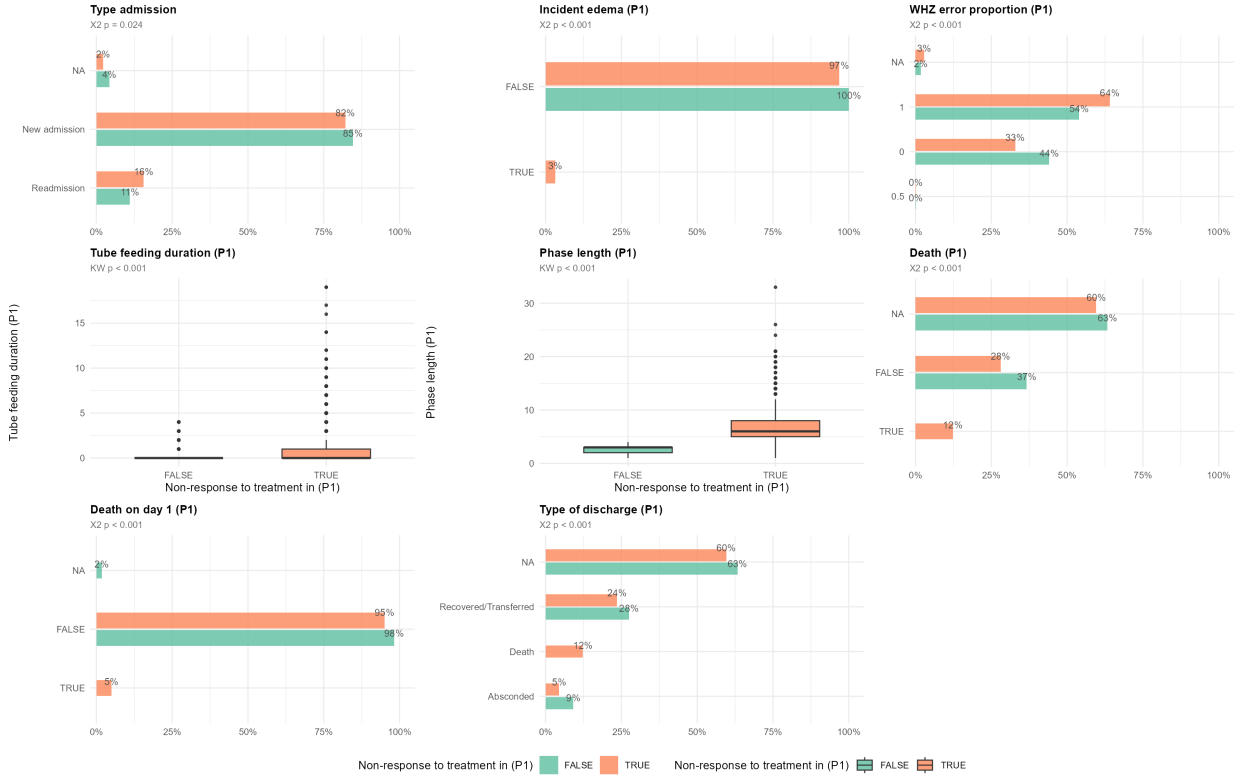
[3] *p<0.05; **p<0.01; ***p<0.001

[4] False discovery rate correction for multiple testing

Sociodemographic



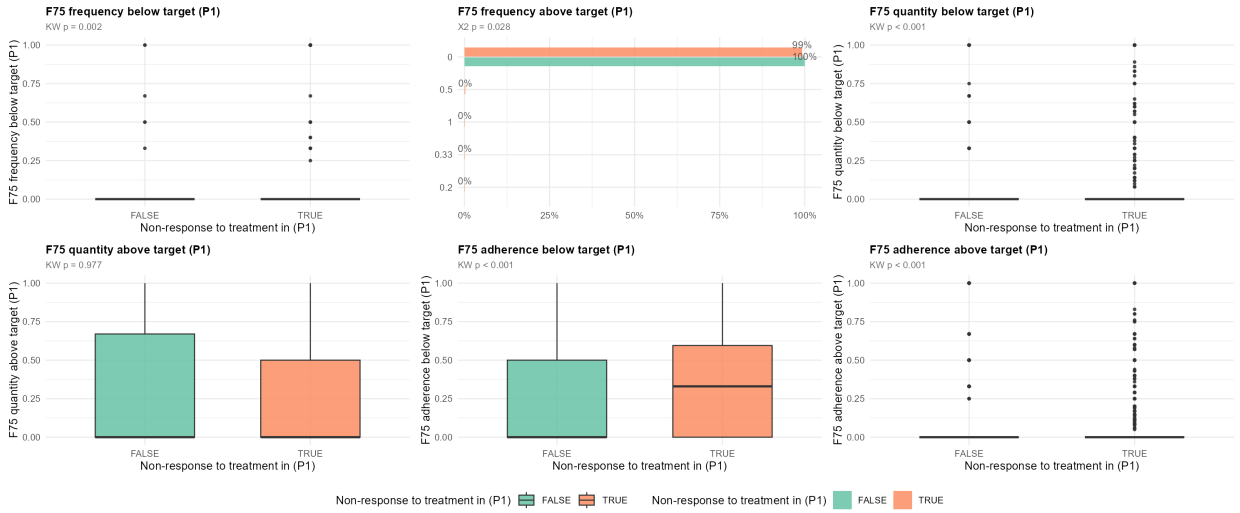
Clinical Admission and Severity



Nutritional Treatment

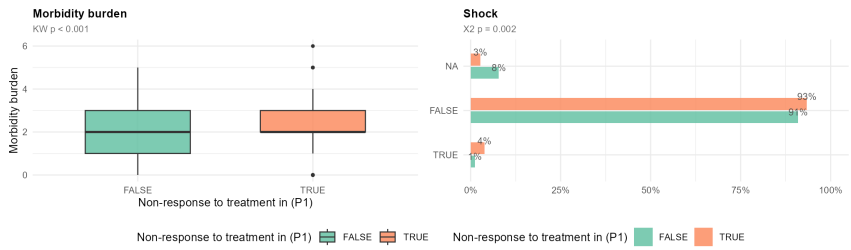


Feeding Compliance

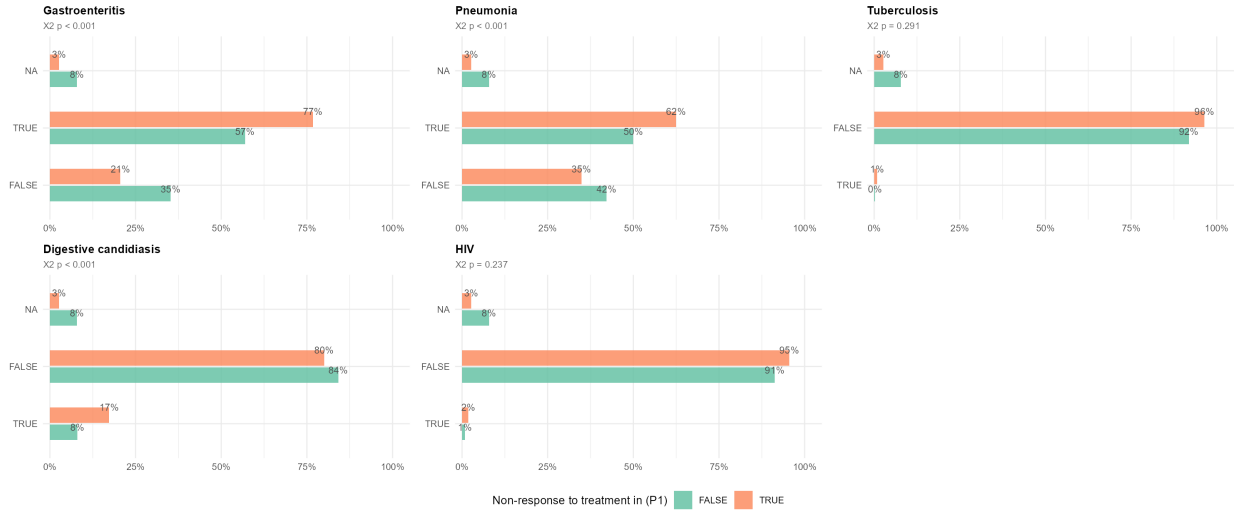


Morbidity and Comorbidities

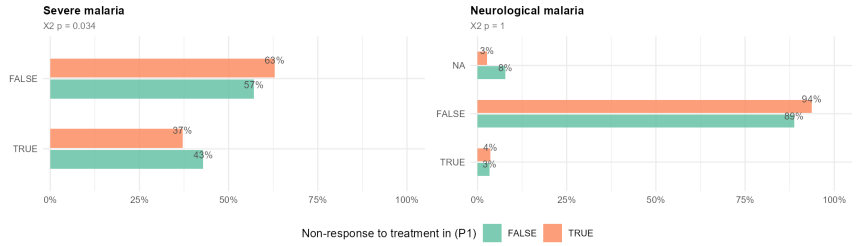
General



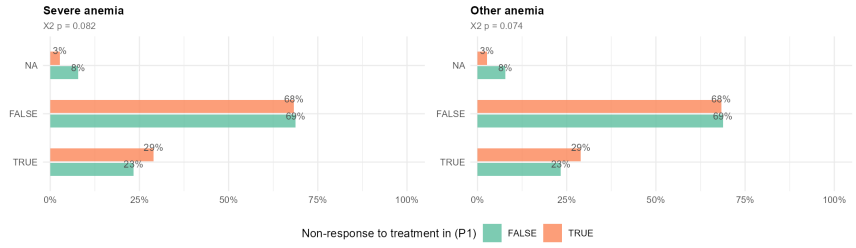
Infections and Parasitic



Malaria



Anaemia



Non-response was strongly associated with markers of treatment intensity and clinical severity rather than with baseline demographic characteristics. Age (p=0.051), age group (p=0.354), sex (p=0.935), and vaccination status (p=0.495) showed no significant association with non-response. In contrast, locality (p=0.004), admission type (p=0.022), incident edema (0% vs. 3.2%, p<0.001), WHZ error proportion (p<0.001), tube feeding duration (median 0 vs. 0, but max 4 vs. 19 days, p<0.001), and phase length (median 3 vs. 6 days, p<0.001) all differed significantly between responders and non-responders. Nutritional treatment pathway was strongly associated with outcome: non-responders were far more likely to have undergone F75-F75F/F75F-F75 transitions (43.1% vs. 7.4%) and F75-to-F75F transition specifically (36.0% vs. 0%, p<0.001), while responders were more often managed on F75 alone (55.0% vs. 29.4%).

Death in phase occurred exclusively among non-responders (30.5% vs. 0%, p<0.001), as did day-1 mortality (5.0% vs. 0%, p<0.001). Feeding-target deviations (F75 frequency and quantity below/above target, F75 adherence below/above target) were all significantly associated with non-response (p 0.038 for most, p<0.001 for adherence measures), except F75 quantity above target (p=0.789). Morbidity burden was higher among non-responders (median 2 [2-3] vs. 2 [1-3], p<0.001), driven by higher prevalence of shock (4.0% vs. 1.3%, p=0.003), gastroenteritis (78.9% vs. 61.8%, p<0.001), pneumonia (64.2% vs. 54.3%, p<0.001), and digestive candidiasis (17.7% vs. 8.7%, p<0.001). Severe malaria showed an inverse association, being more common among responders (42.9% vs. 37.1%, p=0.030). Tuberculosis, HIV, neurological malaria, severe anemia, and other anemia were not significantly associated with non response. Type of discharge differed markedly by outcome status (p<0.001), with death recorded only among non-responders (30.5%) and absconding more frequent among responders (25.0% vs. 11.2%).

Table 2: Results of the univariate logistic regression analysis between explanatory variables and outcome.

Predictor	Event N	OR	95% CI	p-value
F75 adherence below target (P1)				**<0.001**
0	174	—	—	
]0;100]	492	3.07	2.45, 3.87	
Gastroenteritis				**<0.001**
FALSE	137	—	—	
TRUE	511	2.31	1.80, 2.96	
Digestive candidiasis				**<0.001**
FALSE	533	—	—	
TRUE	115	2.27	1.62, 3.23	
F75 adherence above target (P1)				**<0.001**
0	504	—	—	
]0;100]	162	1.90	1.44, 2.51	
Pneumonia				**<0.001**
FALSE	232	—	—	
TRUE	416	1.51	1.21, 1.89	
Shock				**0.002**
FALSE	622	—	—	
TRUE	26	3.21	1.51, 7.65	
Locality				**0.004**
Urban	184	—	—	
Rural	476	0.69	0.54, 0.89	
Type admission				**0.022**
New admission	547	—	—	
Readmission	104	1.45	1.06, 2.00	
Severe malaria				**0.030**
FALSE	419	—	—	

Predictor	Event N	OR	95% CI	p-value
TRUE	247	0.78	0.63, 0.98	
Severe anemia				0.078
FALSE	455	—	—	
TRUE	193	1.25	0.98, 1.60	
Tuberculosis				0.2
FALSE	642	—	—	
TRUE	6	2.90	0.67, 19.9	
HIV				0.2
FALSE	636	—	—	
TRUE	12	1.94	0.75, 5.60	

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

In univariate logistic regression, the strongest predictors of non-response were F75 adherence below target (OR 3.02, 95% CI 2.40-3.81, p<0.001) and shock (OR 3.21, 95% CI 1.51-7.65, p=0.002), followed by gastroenteritis (OR 2.31, 95% CI 1.80-2.96, p<0.001) and digestive candidiasis (OR 2.27, 95% CI 1.62-3.23, p<0.001). Pneumonia (OR 1.51, 95% CI 1.21-1.89, p<0.001) and readmission (OR 1.45, 95% CI 1.06-2.00, p=0.022) were also associated with increased odds of non-response. Rural residence (OR 0.69, 95% CI 0.54-0.89, p=0.004) and severe malaria (OR 0.78, 95% CI 0.63-0.98, p=0.030) were associated with reduced odds. Severe anemia, breastfeeding status, HIV, age group, vaccination status, and sex were not significant univariate predictors ($p \geq 0.078$).

5. Multivariate logistic regression analyses

Table 3 : Results of the generalize logistic regression analysis at Phase 1.

Predictor	N	Event N	OR[1]	95% CI	p-value	q-value[2]	VIF
F75 adherence below target (P1)	1,215		***		<0.001	**<0.001**	1.0
0		154		—			
]0;100]		473	3.16	2.45, 4.09			
F75 adherence above target (P1)	1,215		***		<0.001	**<0.001**	1.0
0		470		—			
]0;100]		157	2.17	1.59, 2.98			
Gastroenteritis	1,215		***		<0.001	**<0.001**	1.1
FALSE		133		—			
TRUE		494	1.82	1.38, 2.40			
Pneumonia	1,215		**		0.003	**0.005**	1.0
FALSE		224		—			
TRUE		403	1.45	1.14, 1.86			
Digestive candidiasis	1,215		*		0.011	**0.015**	1.0
FALSE		514		—			
TRUE		113	1.61	1.11, 2.34			
Type admission	1,215				0.056	**0.065**	1.0
New admission		526		—			
Readmission		101	1.41	0.99, 2.02			
Severe malaria	1,215				0.090	**0.090**	1.0
FALSE		388		—			
TRUE		239	0.81	0.63, 1.03			

[1]*p<0.05; **p<0.01; ***p<0.001

[2]False discovery rate correction for multiple testing

Abbreviations: CI = Confidence Interval, OR = Odds Ratio, VIF = Variance Inflation Factor

In the adjusted GLM model (n=1,203), five predictors retained independent association with non-response after multiple-testing correction, all with low collinearity (VIF=1.0 for all variables). F75 adherence below target remained the strongest independent predictor (adjusted OR 2.89, 95% CI 2.25-3.72, q<0.001), followed by gastroenteritis (aOR 1.89, 95% CI 1.44-2.48, q<0.001), pneumonia (aOR 1.48, 95% CI 1.15-1.89, q=0.004), and digestive candidiasis (aOR 1.64, 95% CI 1.14-2.38, q=0.011). Locality and admission type, both significant in bivariate analysis, were attenuated after adjustment and no longer reached significance (rural aOR 0.78, 95% CI 0.59-1.03, q=0.077; readmission aOR 1.37, 95% CI 0.97-1.95, q=0.077), though

both showed directionally consistent trends with their univariate estimates. Severe malaria, significant and protective in univariate analysis, was not retained in the final multivariable model.